

NETWORKED CONTROL

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1) Consider the continuous-time system

$$\begin{cases} \dot{x}_1 = 4u \\ \dot{x}_2 = \ln(0.5) x_2 + 4 \ln(0.5) u \end{cases}$$

$$y = x_1$$

- A. Assuming that the system is controlled using a **continuous-time control law** of the type $u(t) = Ky(t)$, compute the range of values of K that make the closed-loop system asymptotically stable.
- B. Assuming that the system is controlled using a **digital control law** (i.e., with a sample-and-hold actuator) of the type $u_k = Ky_k$ and with sampling time $h = 1$ s, compute the range of values of K that make the closed-loop system asymptotically stable.
- C. Assume that the system is controlled using the **networked control system** sketched below.

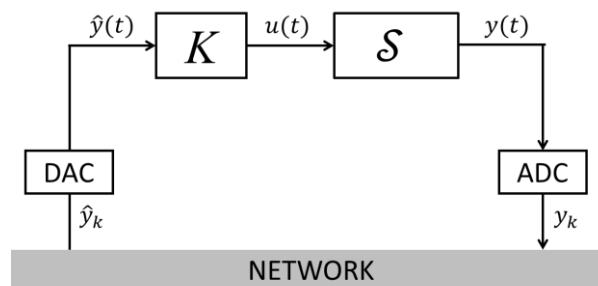


Figure 1: Networked control system

Assume that the sampling time is constant and equal to $h = 1$ s. Define the range of values of K that make the closed-loop system asymptotically stable in case the **network delay is constant**, i.e., $\tau = 0.5$ s.

- D. Assume again that the system is controlled using the networked control system sketched above, with constant sampling time and gain $K = -1/4$. Compute the **maximum allowable transmission interval** in the following two cases:
 - a. $\tau = 0$ s
 - b. $\tau = 0.5$ s

2) Define in details the **delays in the network channel**, commenting the equations that allow to define the propagation and frame time.

Consider a common telephone line (ADSL), where the communication link is a low-pass filter with bandwidth equal to 1 MHz, with signal-to-noise ratio equal to 40 dB. What is the **maximum capacity**?

Assume that the used bitrate is 1Mbits/s, the link has length 100 m and the propagation speed is 2×10^8 m/s. What is the **total delay in the control channel** if we need to transmit a 100-bits packet?

3) Consider a networked control system with an ideal communication channel (i.e., no delays or packet dropouts).

A. In case a digital quantized controller is used, where the scalar input variable u_k is transmitted over a communication channel with rate R and in packets of length N_B bits, and where the system eigenvalues are $\lambda_1, \dots, \lambda_n$, what is the main tradeoff to satisfy in the choice of N_B and of the sampling time h ? Motivate adequately the response.

B. In case a digital quantized controller is used to control the system

$$\begin{cases} \dot{x}_1(t) = 3x_1(t) - x_2(t) + 2u(t) \\ \dot{x}_2(t) = 3x_2(t) - x_1(t) \\ y(t) = x_1(t) \end{cases}$$

and with sampling time $h = 0.5$ s, what is the minimum number of bits N_B encoding the control input required to make the system boundable?